

Ejdo II

$$x dx - \frac{\sqrt{4-x}}{y} dy = 0$$

$$\frac{x dx}{\sqrt{4-x}} - y^{-1} dy = 0$$

$$\int \frac{x}{\sqrt{4-x}} - \int \frac{dy}{y} = \int 0$$

$$-8\sqrt{4-x^2} + \frac{2}{3}(4-x^2)\sqrt{4-x^2} + \ln y = h$$

$$t = \sqrt{4-x}$$

$$x = 4-t^2$$

$$dx = -2t dt$$

$$- \int \frac{4-t^2}{t} \cdot 2t dt$$

$$= - \int 8 - 2t^2 dt$$

$$= -8t + \frac{2}{3}t^3 \rightarrow -8\sqrt{4-x^2} + \frac{2}{3}(\sqrt{4-x^2})^3$$

Halla la solución particular de:
 $(1+x^3)dy - x^2 y dx = 0$ que satisfaga
 las condiciones iniciales, $x = 1; y = 2$.

$$\frac{(1+x^3) dy}{(1+x^3)} - \frac{x^2 y dx}{(1+x^3)} = 0$$

$$dy - \frac{x^2 y dx}{(1+x^3)} = 0$$

$$\frac{dy}{y} - \frac{x^2 dx}{(1+x^3)} = 0$$

$$\int \frac{dy}{y} - \int \frac{x^2}{(1+x^3)} dx = \int 0$$

$$\ln |y| - \frac{1}{3} \ln |1+x^3| = \ln h$$

$$3 \ln |y| - \ln |1+x^3| = 3 \ln h$$

$$3 \ln |y| - \ln |1+x^3| = \ln c$$

$$\ln |y|^3 = \ln |1+x^3| + \ln c$$

$$\ln |y^3| = \ln [c(1+x^3)]$$

$$y^3 = c(1+x^3)$$

$$x = 1, y = 2$$

$$t = 1+x^3$$

$$dt = 3x^2 dx$$

$$\frac{dt}{3x^2} = dx$$

$$\int \frac{x^2 dt}{t \cdot 3x^2}$$

$$\frac{1}{3} \int \frac{dt}{t} = \frac{1}{3} \ln t = \frac{1}{3} \ln |1+x^3|$$

$$(1+x^3)dy - x^2 y dx = 0$$

$$(1+x^3)dy = x^2 y dx$$

$$\frac{dy}{dx} = \frac{x^2 y}{(1+x^3)}$$

$$2^3 = C(1+1)$$

$$8 = 2C$$

$$C = 4$$

$$y^3 = 4(1+x^3)$$

$$3y^2 \cdot y' = 4 + 4x^3$$

$$y' = \frac{4 + 4x^3}{3y^2}$$

$$y^3 = \frac{4(1+x^3)}{3y^2}$$

$$\frac{dy}{dx} = 3x - 1$$

$$dy = 3x dx - dx$$

$$\int dy = \int 3x dx - \int dx$$

$$\int dy - \int 3x dx + \int dx = 0$$

$$y - \frac{3x^2}{2} + x = k$$

$$2y - 3x^2 + 2x = 2k$$

$$2y + 2x - 3x^2 = 2k$$

$$2y + x(2 - 3x) = 2k$$

$$2y + x(2 - 3x) = C$$

$$2) ydx - xdy = 0$$

$$\frac{dx}{x} - \frac{dy}{y} = 0$$

$$\ln|x| - \ln|y| = \ln k$$

$$\ln\left|\frac{x}{y}\right| = \ln k$$

$$\frac{x}{y} = k$$

$$3) \frac{dy}{dx} + y \cos x = 0$$

$$dy = -y \cos x dx$$

$$dy + y \cos x dx = 0$$

$$\frac{dy}{y} + \cos x dx = 0$$

$$\ln|y| + \sin x = k$$

$$\ln|y| = k - \sin x$$

$$y = e^{k - \sin x}$$

$$4) (x-1)dy - ydx = 0$$

$$\frac{dy}{y} - \frac{dx}{(x-1)} = 0$$

$$\ln|y| - \ln|x-1| = \ln k$$

$$\ln|y| = \ln k + \ln|x-1|$$

$$\ln|y| = \ln(k(x-1))$$

$$y = k(x-1)$$

$$5) (4y + yx^2)dy - (2x + xy^2)dx = 0$$

$$y(4+x^2)dy - x(2+y^2)dx = 0$$

$$\int \frac{y dy}{2+y^2} - \int \frac{x dx}{4+x^2} = \int 0$$

$$\frac{1}{2} \ln|2+y^2| - \frac{1}{2} \ln|4+x^2| = \ln c$$

$$\ln|2+y^2| = \ln c^2 + \ln|4+x^2|$$

$$\ln|2+y^2| = \ln c^2(4+x^2)$$

$$2+y^2 = c^2(4+x^2)$$

$$2+y^2 = k(4+x^2)$$

$$t = 2+y^2$$

$$dt = 2y$$

$$\int \frac{y dt}{t \cdot 2y}$$

$$u = 4+x^2$$

$$du = 2x$$

$$\int \frac{du}{u}$$

$$6) \sin x(e^{-y} + 1)dx = (1 + \cos x)dy; x=0, y=0$$

$$\sin x(e^{-y} + 1)dx - (1 + \cos x)dy = 0$$

$$\sin x(e^{-y} + 1)dx - 1 dy - \cos x dy = 0$$

